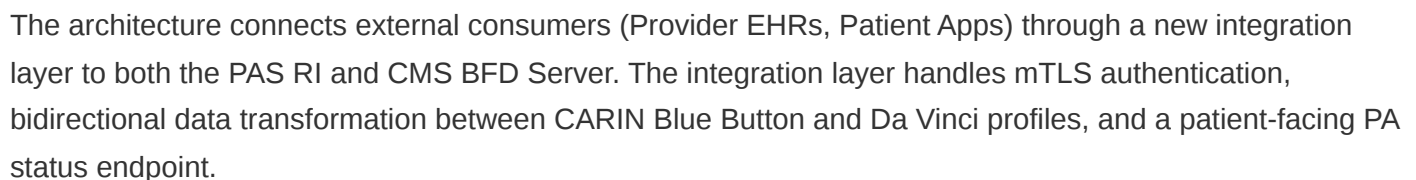
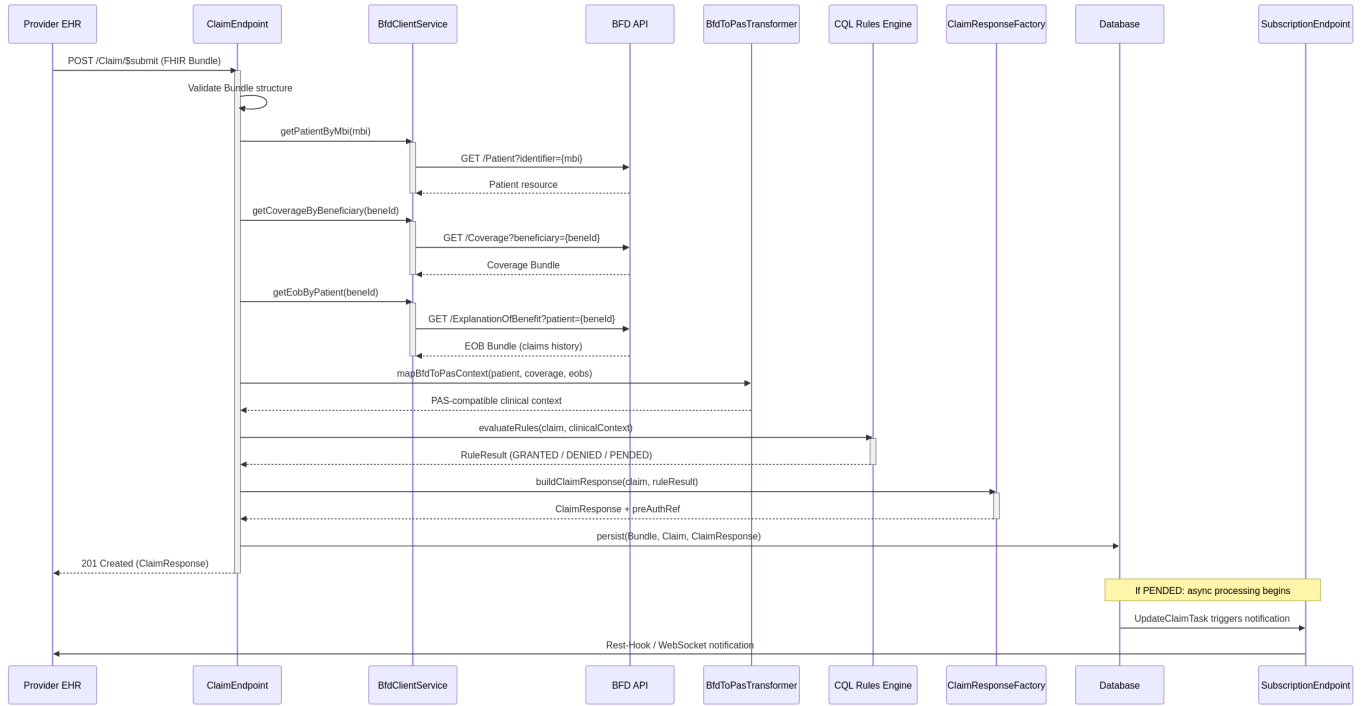


Executive Summary

1. Architecture Overview

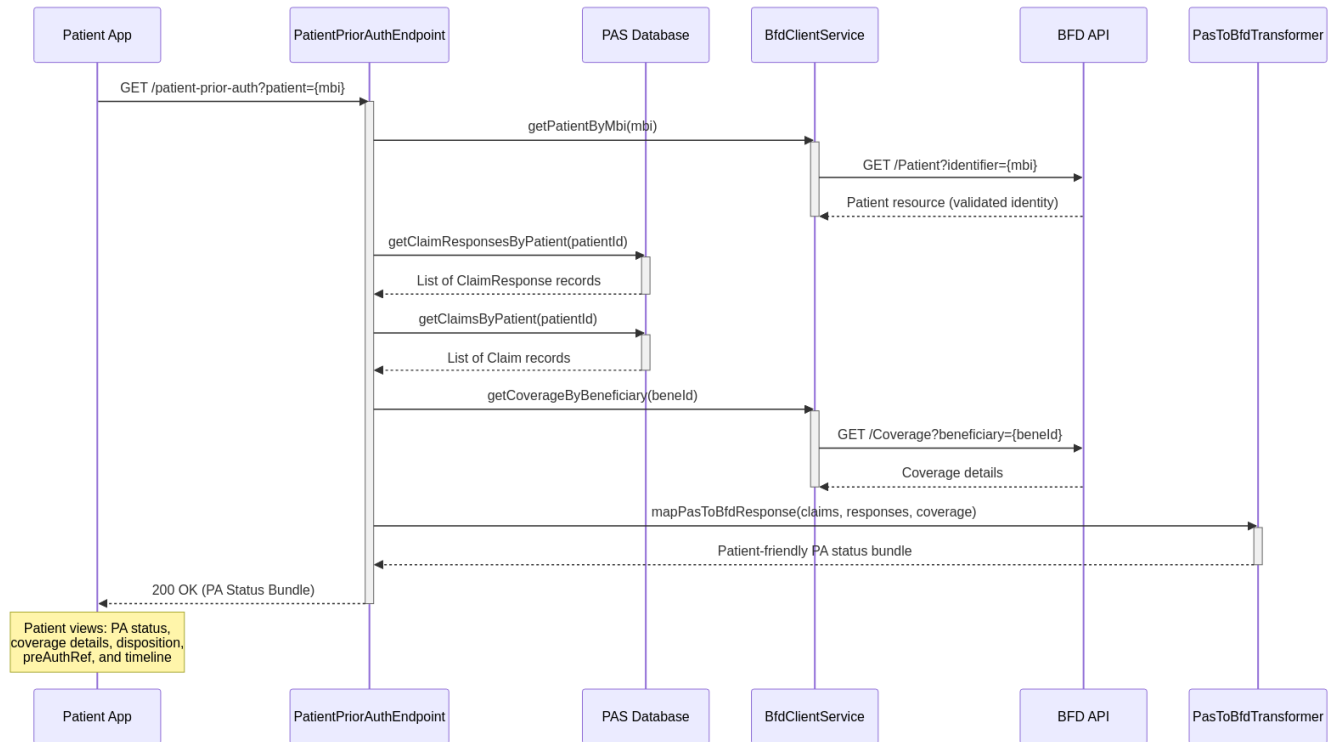


2. Claim Submission Flow (BFD-Enriched)



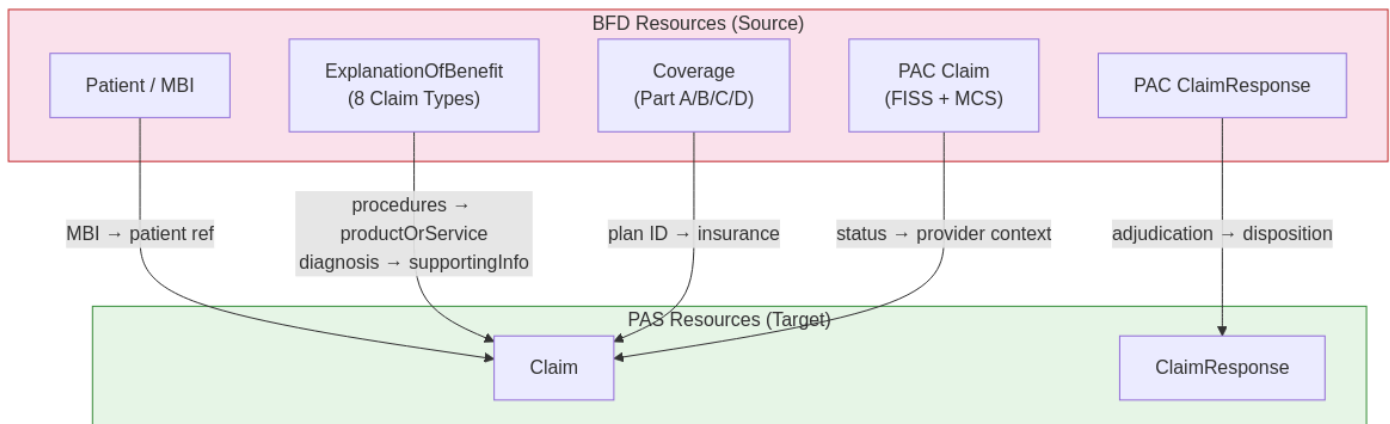
When a provider submits a prior authorization request, the system validates the patient's MBI against BFD, retrieves coverage and claims history, transforms the data into PAS-compatible structures, and evaluates CQL clinical rules before generating a ClaimResponse with disposition (Granted / Denied / Pended).

3. Patient Access to PA Status



Patient-facing applications retrieve prior authorization status through the PatientPriorAuthEndpoint, which combines PAS database records with BFD beneficiary data — satisfying the CMS-0057-F Patient Access API requirement.

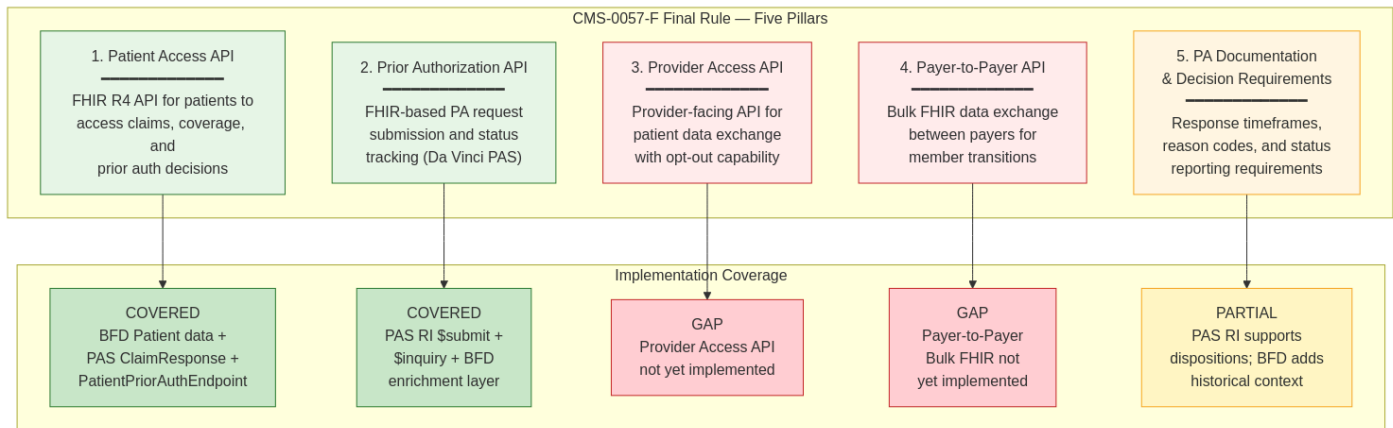
4. Field-Level Data Mapping (BFD to PAS)



BFD resources (Patient/MBI, ExplanationOfBenefit across 8 claim types, Coverage, PAC Claims FISS/MCS, PAC ClaimResponses) are mapped to PAS Claim and ClaimResponse fields through dedicated transformer

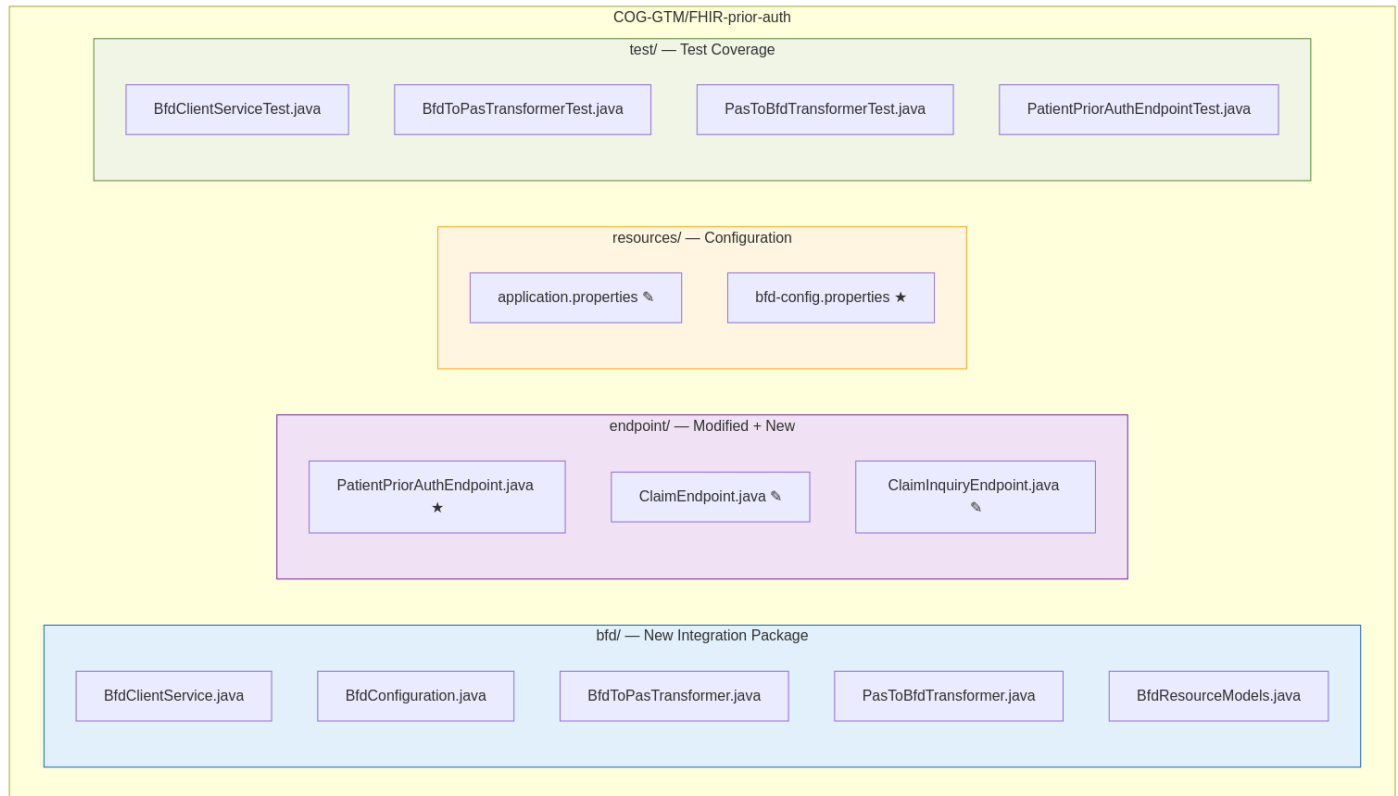
classes. Key mappings include MBI to patient reference, EOB procedure codes to item.productOrService, Coverage to insurance, and PAC adjudication to ClaimResponse disposition.

5. CMS-0057-F Compliance Coverage



Pillar	Status	Notes
Patient Access API	Covered	PatientPriorAuthEndpoint + BFD Patient/Coverage/EOB
Prior Authorization API	Covered	PAS RI \$submit/\$inquiry + BFD enrichment
Provider Access API	Gap	Requires new endpoint + consent management
Payer-to-Payer API	Gap	Requires Bulk FHIR + inter-payer trust framework
PA Documentation	Partial	PAS dispositions supported; SLA enforcement + X12 278 reason codes needed

6. Module Layout



The integration adds 13 new files (5 in `bfd/`, 1 new endpoint, 1 config file, 4 tests, 2 doc/script files) and modifies 7 existing files (ClaimEndpoint, ClaimInquiryEndpoint, App, Database, FhirUtils, application.properties, build.gradle).

Prepared for CMS-0057-F compliance readiness review.